"BOILERS"

* **Introduction :-** Boiler or "steam generator" is closed vessel in which water is converted into steam by application of heat is provided by combustion of fuel.

→ closed vessel which has a volume more than 22.75 litr.

→ Pressure must be more than $3.5\ kg/cm^2 \simeq 50\ PSI$

→ It has the attachments like mountings & fittings.

* **Classification of Boiler**

1) **Based on Tube Contain :-**

a) fire Tube Boiler

b) Water Tube Boiler

2) Position of furnace :-

a) Externally fired Boiler

b) Internally fired Boiler

3) According to Position of Boiler

a) Vertical

b) Horizontal

c) Inclined

4) According to the use :-

a) Stationary Boiler.

b) Portable Boiler or Mobile Boiler.

5) According to the water & steam circulation

a) Natural Circulation

b) forced Circulation

6) On the Basis of ~~the~~ fuel used :-

a) Coal fired Boiler.

b) Oil fired Boiler.

c) Gas " "

d) Electrically heated Boiler

e) Nuclear heat Boiler

f) Solar Energy.

g) Hot ~~waste~~ waste Gases.

7) According to Pressure Developed :-

a) Low Pressure Boiler

b) High Pressure Boiler.

I) Sub-Critical Boiler.

II) Super " "

8) Based on fuel Combustion System :-

a) suspension Combustion

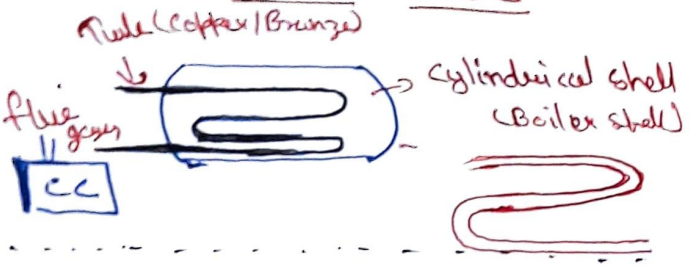
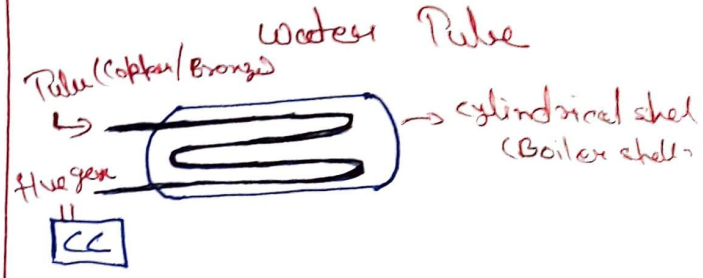
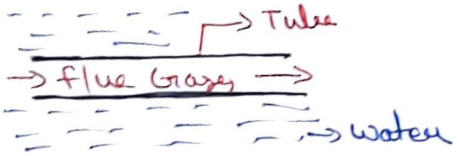
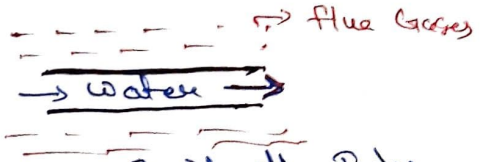
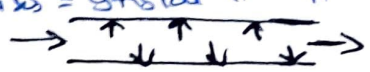
b) Bed Combustion

9) Based on fuel feeding :-

(a) Above Bed.

(b) Under Bed.

* Difference B/w fire & water Pulse Boiler :-

fire Pulse	water Pulse
	
	
→ water = outside the Tube → flue Gases = inside " " 	→ water = inside the Tube → Flue Gases = outside " "
σ_c = Circumferential stress $\sigma_c = \frac{Pd}{2t}$, $P\uparrow \rightarrow \sigma_c\uparrow$	
→ Low Pressure [upto 20 bar/sq.in]	→ High Pressure / Low Press. (above 50 bar)
→ Natural Circulation (ρ difference)	→ forced Circulation (Pump).
→ Internally fired Preferred (CC inside the Boiler shell)	→ Externally fired (CC outside the Boiler shell).
→ Size: Larger, for the same amount of steam Production	→ Size: Smaller Size, for the same amount of steam Production
→ η = 75 % (upto)	→ η = 90% (upto)
→ occupies Larger space Area	→ smaller space area.

Steam Boilers :- Important Terminology

1) **Boiler shell** :- This cylindrical shell is formed by welding or riveting the steel Plates whose ends are sealed by end Plates. It is used to store the water to be heated. Also Its contains the steam that is formed as a result of heating. Therefore, it should have sufficient storage capacity

2) **Combustion chamber** :- The combustion chamber, usually located below the boiler shell is used to burn fuel so that water in the boiler shell can be heated to produce the steam.

3) **Grate** :- A Grate acts as a platform inside the combustion chamber to burn the coal or wood fuel. It is made from cast Iron bars, the space b/w which allows the air to Pass for Combustion. Total Surface area of the grate on which Combustion takes Place is called Grate Surface

4) **furnace** :- It is located b/w the grate and Boiler shell. The fuel is actually burn here. It also known as fire box.

5) **Heating Surface** :- The Portion of the boiler Surface exposed to fire and hot Gases is called heating Surfaces.

6.) <u>Mountings</u> :- Certain components are fitted on the boiler to ensure proper functioning and safety. These are called mountings. for Exam) - Safety Valve, water level indicator pressure gauge etc.

7) <u>Accessories</u> :- These are the additional devices used with the boiler without mountings them on it. These are used to increase the efficiency of the boiler. for Example :- superheater, economizer, feed pump etc.